UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which declaration creates an integer array of five elements?
   A. int a(5);
   B. int a[5];
   C. int a;
   D. array a[5];

2. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. n
   B. t
   C. p
   D. i

3. Which operator reads data from cin?
   A. =
   B. >>
   C. %
   D. <<

4. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 2468
   B. 246
   C. 1234
   D. 1357

5. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. ii and iv
   B. i and ii
   C. iii and iv
   D. i and iii

6. What does a compiler do?
   A. Stores only integer values
   B. Draws flowcharts automatically
   C. Translates the whole source program before execution
   D. Executes one source statement at a time

7. Which declaration creates a floating-point variable named price?
   A. float price;
   B. char price;

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C. int price;
D. bool price;

8. Which term describes a finite sequence of precise steps that solves a problem?
A. Compiler
B. Array
C. Algorithm
D. Variable

9. Which of the following are selection structures?
i. if
ii. switch
iii. for
iv. while
A. i and iii
B. ii and iv
C. i and ii
D. iii and iv

10. How many times does a while loop execute when its condition is false initially?
A. Until the user stops it
B. Zero
C. Two
D. One

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(3 marks)

b) Explain TWO advantages of high-level languages over machine language.

(2 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(7 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
    total += i;
}
cout << total;
```

a) Trace the program and state the output.

(7 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(8 marks)

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QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(8 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A catering company charges the meal prices shown in Table 1.

Table 4. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

END OF QUESTION PAPER

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